

4.6.4 Data Dictionary

The following data dictionary documents the twenty-one tables implemented in the PostgreSQL database of the Cross-Institutional Academic Collaboration and Real-Time Event Notification System. It is based directly on the supplied system schema. Each table identifies the stored fields, PostgreSQL data types, null rules, keys, constraints, default values and the purpose of each field.

Notation: PK = Primary Key; FK = Foreign Key; UQ = Unique constraint; TIMESTAMPTZ = timestamp with time zone. A column marked “Yes” under Nullable accepts NULL according to the implemented SQL schema.

Table 4.6.4-1: Data Dictionary for institutions

Purpose: Stores the higher-education and research institutions registered in the platform.

Field	Data type	Null?	Key / constraint	Default	Description
id	SERIAL	No	PK	Auto-generated	Unique institution identifier.
name	VARCHAR(255)	No	UNIQUE	None	Official institution name.
type	VARCHAR(100)	No	-	None	Institution category, such as University or Research Institute.
location	VARCHAR(255)	No	-	None	Institution's physical location.
created_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the institution record was created.

Table 4.6.4-2: Data Dictionary for departments

Purpose: Stores departments and links each department to one institution.

Field	Data type	Null?	Key / constraint	Default	Description
id	SERIAL	No	PK	Auto-generated	Unique department identifier.
name	VARCHAR(255)	No	UQ with institution_id	None	Department name; unique within its institution.
institution_id	INT	No	FK -> institutions.id; ON DELETE CASCADE	None	Institution that owns the department.
created_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the department was created.

Table 4.6.4-3: Data Dictionary for users**Purpose:** Stores user accounts, academic roles, affiliations and profile information.

Field	Data type	Null?	Key / constraint	Default	Description
id	SERIAL	No	PK	Auto-generated	Unique user identifier.
email	VARCHAR(255)	No	UNIQUE	None	User's sign-in email address.
password_hash	VARCHAR(255)	No	-	None	Secure hash of the user's password.
full_name	VARCHAR(255)	No	-	None	User's full name.
role	VARCHAR(50)	No	CHECK: student, lecturer, researcher, admin	None	Authorization role assigned to the user.
institution_id	INT	Yes	FK -> institutions.id; ON DELETE SET NULL	NULL	Institution with which the user is affiliated.
department_id	INT	Yes	FK -> departments.id; ON DELETE SET NULL	NULL	Department with which the user is affiliated.
bio	TEXT	Yes	-	NULL	Short academic or professional biography.
avatar_url	VARCHAR(255)	Yes	-	NULL	Path or URL of the user's profile image.
status	VARCHAR(50)	Yes	CHECK: active, suspended	'active'	Current account status.
created_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the account was created.

Table 4.6.4-4: Data Dictionary for academic_communities**Purpose:** Stores public, private or institution-specific academic communities.

Field	Data type	Null?	Key / constraint	Default	Description
id	SERIAL	No	PK	Auto-generated	Unique community identifier.
name	VARCHAR(255)	No	UNIQUE	None	Community name.
description	TEXT	Yes	-	NULL	Purpose and scope of the community.
category	VARCHAR(100)	No	-	None	Academic category or subject area.
created_by	INT	Yes	FK -> users.id; ON DELETE SET NULL	NULL	User who created the community.
institution_id	INT	Yes	FK -> institutions.id; ON DELETE SET NULL	NULL	Owning institution; NULL indicates a cross-institutional community.
created_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the community was created.
privacy_type	VARCHAR(50)	Yes	CHECK: public, private, institution	'public'	Controls who may discover or access the community.

Table 4.6.4-5: Data Dictionary for community_members**Purpose:** Links users to the academic communities they have joined.

Field	Data type	Null?	Key / constraint	Default	Description
community_id	INT	No	PK; FK -> academic_communities.id; ON DELETE CASCADE	None	Community membership identifier.
user_id	INT	No	PK; FK -> users.id; ON DELETE CASCADE	None	User membership identifier.
joined_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the user joined.

Table 4.6.4-6: Data Dictionary for community_invitations**Purpose:** Stores invitations issued to users for private or restricted communities.

Field	Data type	Null?	Key / constraint	Default	Description
id	SERIAL	No	PK	Auto-generated	Unique invitation identifier.
community_id	INT	Yes	FK -> academic_communities.id; ON DELETE CASCADE; UQ with user_id	NULL	Community to which the user is invited.
user_id	INT	Yes	FK -> users.id; ON DELETE CASCADE; UQ with community_id	NULL	Invited user.
status	VARCHAR(50)	Yes	CHECK: pending, accepted, rejected	'pending'	Current invitation decision.
created_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the invitation was created.

Table 4.6.4-7: Data Dictionary for posts**Purpose:** Stores discussion posts published inside academic communities.

Field	Data type	Null?	Key / constraint	Default	Description
id	SERIAL	No	PK	Auto-generated	Unique post identifier.
title	VARCHAR(255)	No	-	None	Post title.
content	TEXT	No	-	None	Main post content.
user_id	INT	No	FK -> users.id; ON DELETE CASCADE	None	User who authored the post.
community_id	INT	No	FK -> academic_communities.id; ON DELETE CASCADE	None	Community in which the post was published.
created_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the post was created.

Table 4.6.4-8: Data Dictionary for comments**Purpose:** Stores user comments attached to community posts.

Field	Data type	Null?	Key / constraint	Default	Description
id	SERIAL	No	PK	Auto-generated	Unique comment identifier.
post_id	INT	No	FK -> posts.id; ON DELETE CASCADE	None	Post receiving the comment.
user_id	INT	No	FK -> users.id; ON DELETE CASCADE	None	User who wrote the comment.
content	TEXT	No	-	None	Comment text.
created_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the comment was created.

Table 4.6.4-9: Data Dictionary for likes**Purpose:** Records which users have liked which community posts.

Field	Data type	Null?	Key / constraint	Default	Description
post_id	INT	No	PK; FK -> posts.id; ON DELETE CASCADE	None	Liked post identifier.
user_id	INT	No	PK; FK -> users.id; ON DELETE CASCADE	None	User who liked the post.
created_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the like was recorded.

Table 4.6.4-10: Data Dictionary for projects**Purpose:** Stores private academic collaboration projects and their access rules.

Field	Data type	Null?	Key / constraint	Default	Description
id	SERIAL	No	PK	Auto-generated	Unique project identifier.
title	VARCHAR(255)	No	-	None	Project title.
description	TEXT	No	-	None	Detailed project description.
created_by	INT	Yes	FK -> users.id; ON DELETE SET NULL	NULL	User who created the project.
status	VARCHAR(50)	Yes	CHECK: planning, active, completed	'planning'	Current project lifecycle status.
requirements	JSONB	No	CHECK: JSON value is an array	'[]::jsonb	Structured list of skills or participation requirements.
privacy_type	VARCHAR(20)	No	CHECK: private only	'private'	Project privacy mode.
access_scope	VARCHAR(50)	No	CHECK: everyone, institution	'everyone'	Determines whether requests may come from everyone or only the institution.
institution_id	INT	Yes	FK -> institutions.id; ON DELETE SET NULL	NULL	Institution used when access is institution-restricted.
updated_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time of the latest project update.
created_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the project was created.

Table 4.6.4-11: Data Dictionary for project_members**Purpose:** Links users to projects and records their participation role.

Field	Data type	Null?	Key / constraint	Default	Description
project_id	INT	No	PK; FK -> projects.id; ON DELETE CASCADE	None	Project membership identifier.
user_id	INT	No	PK; FK -> users.id; ON DELETE CASCADE	None	Participating user identifier.
role	VARCHAR(50)	Yes	CHECK: lead, contributor, observer	'contributor'	User's responsibility within the project.
joined_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the user joined the project.

Table 4.6.4-12: Data Dictionary for project_join_requests**Purpose:** Stores user requests to join private academic projects.

Field	Data type	Null?	Key / constraint	Default	Description
id	SERIAL	No	PK	Auto-generated	Unique join-request identifier.
project_id	INT	No	FK -> projects.id; ON DELETE CASCADE; UQ with user_id	None	Project the user wants to join.
user_id	INT	No	FK -> users.id; ON DELETE CASCADE; UQ with project_id	None	User requesting project membership.
status	VARCHAR(50)	No	CHECK: pending, accepted, rejected	'pending'	Current decision on the request.
requested_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the request was submitted.
responded_at	TIMESTAMPTZ	Yes	-	NULL	Date and time when the request was decided.
responded_by	INT	Yes	FK -> users.id; ON DELETE SET NULL	NULL	User who accepted or rejected the request.

Table 4.6.4-13: Data Dictionary for project_files**Purpose:** Stores metadata for files uploaded to collaboration projects.

Field	Data type	Null?	Key / constraint	Default	Description
id	SERIAL	No	PK	Auto-generated	Unique project-file identifier.
project_id	INT	No	FK -> projects.id; ON DELETE CASCADE	None	Project that owns the file.
filename	VARCHAR(255)	No	-	None	Original or displayed file name.
filepath	VARCHAR(255)	No	-	None	Server-side storage path.
mime_type	VARCHAR(255)	Yes	-	NULL	File's media type.
file_size	BIGINT	Yes	-	NULL	File size in bytes.
uploaded_by	INT	Yes	FK -> users.id; ON DELETE SET NULL	NULL	User who uploaded the file.
uploaded_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the file was uploaded.

Table 4.6.4-14: Data Dictionary for events**Purpose:** Stores academic events, schedules, venues and participation capacity.

Field	Data type	Null?	Key / constraint	Default	Description
id	SERIAL	No	PK	Auto-generated	Unique event identifier.
title	VARCHAR(255)	No	-	None	Event title.
description	TEXT	Yes	-	NULL	Detailed event information.
event_date	TIMESTAMPTZ	No	-	None	Scheduled date and time of the event.
location	VARCHAR(255)	No	-	None	Physical venue or location description.
meeting_link	VARCHAR(1000)	Yes	-	NULL	Online meeting URL, when applicable.
organizer_id	INT	Yes	FK -> users.id; ON DELETE SET NULL	NULL	User responsible for organizing the event.
institution_id	INT	Yes	FK -> institutions.id; ON DELETE SET NULL	NULL	Host institution; NULL means the event is open across institutions.
capacity	INT	Yes	-	100	Maximum planned number of participants.
updated_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time of the latest event update.
created_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the event was created.

Table 4.6.4-15: Data Dictionary for event_registrations**Purpose:** Links users to the academic events for which they have registered.

Field	Data type	Null?	Key / constraint	Default	Description
event_id	INT	No	PK; FK -> events.id; ON DELETE CASCADE	None	Registered event identifier.
user_id	INT	No	PK; FK -> users.id; ON DELETE CASCADE	None	Registered user identifier.
registered_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when registration occurred.

Table 4.6.4-16: Data Dictionary for news**Purpose:** Stores public academic news and announcements.

Field	Data type	Null?	Key / constraint	Default	Description
id	SERIAL	No	PK	Auto-generated	Unique news item identifier.
title	VARCHAR(255)	No	-	None	News headline.
description	TEXT	No	-	None	News content or summary.
category	VARCHAR(50)	No	-	'Other'	News category.
feature_image	VARCHAR(500)	No	-	None	Path or URL of the main news image.
external_link	VARCHAR(1000)	Yes	-	NULL	Optional link to an external source or related page.
created_by	INT	Yes	FK -> users.id; ON DELETE SET NULL	NULL	User who published the news item.
created_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the item was created.
updated_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time of the latest news update.

Table 4.6.4-17: Data Dictionary for news_documents**Purpose:** Stores document attachments associated with public news items.

Field	Data type	Null?	Key / constraint	Default	Description
id	SERIAL	No	PK	Auto-generated	Unique news-document identifier.
news_id	INT	No	FK -> news.id; ON DELETE CASCADE	None	News item that owns the document.
filename	VARCHAR(255)	No	-	None	Original or displayed document name.
filepath	VARCHAR(500)	No	-	None	Server-side storage path.
mime_type	VARCHAR(150)	Yes	-	NULL	Document media type.
file_size	BIGINT	Yes	-	NULL	Document size in bytes.
created_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the document was attached.

Table 4.6.4-18: Data Dictionary for chat_rooms**Purpose:** Stores one-to-one and group chat rooms.

Field	Data type	Null?	Key / constraint	Default	Description
id	SERIAL	No	PK	Auto-generated	Unique chat-room identifier.
name	VARCHAR(255)	Yes	-	NULL	Room name; NULL is used for a one-to-one conversation.
is_group	BOOLEAN	Yes	-	FALSE	Indicates whether the room is a group chat.
created_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the room was created.

Table 4.6.4-19: Data Dictionary for chat_members**Purpose:** Links users to the chat rooms in which they participate.

Field	Data type	Null?	Key / constraint	Default	Description
room_id	INT	No	PK; FK -> chat_rooms.id; ON DELETE CASCADE	None	Chat-room membership identifier.
user_id	INT	No	PK; FK -> users.id; ON DELETE CASCADE	None	Participating user identifier.
joined_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the user joined the room.

Table 4.6.4-20: Data Dictionary for chat_messages**Purpose:** Stores messages sent within chat rooms.

Field	Data type	Null?	Key / constraint	Default	Description
id	SERIAL	No	PK	Auto-generated	Unique message identifier.
room_id	INT	No	FK -> chat_rooms.id; ON DELETE CASCADE	None	Chat room containing the message.
sender_id	INT	No	FK -> users.id; ON DELETE CASCADE	None	User who sent the message.
message	TEXT	No	-	None	Message content.
created_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the message was sent.

Table 4.6.4-21: Data Dictionary for notifications**Purpose:** Stores persistent in-application notifications delivered to individual users.

Field	Data type	Null?	Key / constraint	Default	Description
id	SERIAL	No	PK	Auto-generated	Unique notification identifier.
user_id	INT	No	FK -> users.id; ON DELETE CASCADE	None	User receiving the notification.
title	VARCHAR(255)	No	-	None	Short notification heading.
content	TEXT	No	-	None	Notification message.
type	VARCHAR(50)	No	Application values: chat, event, project, community, system	None	Notification category used by the interface.
is_read	BOOLEAN	Yes	-	FALSE	Indicates whether the user has opened or acknowledged the notification.
link	VARCHAR(255)	Yes	-	NULL	Optional internal destination related to the notification.
created_at	TIMESTAMPTZ	Yes	-	CURRENT_TIMESTAMP	Date and time when the notification was created.